

# PHONETICS AND PHONOLOGY

Rhythm and Meter

1

Edited by

Paul Kiparsky  
Gilbert Youmans

# **PHONETICS AND PHONOLOGY**

**VOLUME 1**

# PHONETICS and PHONOLOGY

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# PHONETICS and PHONOLOGY

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## VOLUME 1

### Rhythm and Meter

Edited by

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## PREFACE

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*Rhythm and Meter* is a collection of original articles by twelve linguists and literary critics who have made important contributions to current theories of phonology, verse meter, and music. The articles grew out of papers presented at an international conference on metrical theory held at Stanford University in 1984. Participants circulated draft copies of their papers before the conference, discussed them at the conference, then revised them in response to suggestions and criticisms offered by the editors and other contributors. Hence, this volume represents a rare collaboration between leading scholars in two disciplines, linguistics and literary criticism—disciplines that tend too often to treat each other as adversaries rather than allies.

Most of the articles in this collection focus on English poetry—on the meter of Shakespeare and other Renaissance dramatists, on Milton, Blake, Longfellow, Hopkins, and Auden; but poetry in other languages is also examined: Greek and Arabic (by Alan Prince), Hebrew (by Morris Halle), Russian (by Marina Tarlinskaja), and German (by Beth Bjorklund). In addition, Ray Jackendoff investigates relationships between rhythm in language and in music.

The emphasis in *Rhythm and Meter* is on metrical theory, on formulating and illustrating metrical principles within the tradition of generative metrics (first developed by Morris Halle and S. Jay Keyser) and within competing traditions. Nevertheless, phonological theory remains a central concern for

most of the authors since metrical principles are necessarily based on phonological ones, and metrical practice often provides evidence for specific phonological theories. Hence, *Rhythm and Meter* is addressed to phonologists as well as to metrical theorists—indeed to all scholars interested in applying linguistic analysis to the study of literature, and vice versa.

The articles in this collection represent a wide variety of scholarly approaches. In his analysis of Greek and Arabic meters, Alan Prince sketches what amounts to a universal generative theory of meter, while Elizabeth Closs Traugott concentrates on a single poem by W. H. Auden. Bruce Hayes treats *Hiawatha* less as an aesthetic object than as a subject for generative phonological analysis, while Edward Weismiller avoids linguistic and metrical formalisms altogether to focus on the rhythmic effects of verse. Paul Kiparsky proposes a new generative theory of sprung rhythm, while Derek Attridge and Marina Tarlinskaja criticize generative metrics and propose alternative metrical theories of their own.

The diversity of these approaches reflects a tendency of the linguists and literary critics represented here to divide into opposing (if not warring) camps. In the early stages of the 1984 Stanford Metrics Conference, the scholars from the two disciplines approached each other warily, and they ate at different restaurants. By the end of the conference, however, they managed to reach an accommodation (if not an agreement). Each group recognized that it had much to learn from the other, and the articles in this collection are best understood as contributions to a continuing dialectic rather than to a debate. Disagreements remain, but a consensus emerges on many important metrical principles. Hence, despite their differing approaches, the articles in *Rhythm and Meter* combine to form a coherent whole rather than a miscellaneous collection.

The 1984 Stanford Metrics Conference was organized by Paul Kiparsky. It was funded through a gift from the System Development Foundation through the Center for the Study of Language and Information, Stanford University. Other participants in the conference included Stephen Anderson, T. V. F. Brogan, and Donca Steriade, who chaired sessions, and Mark Liberman, who presented a paper.

Gilbert Youmans

## INTRODUCTION: RHYTHM AND METER

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Music isn't the universal language;  
rhythm is. Plenty of people are tone deaf,  
but everyone has a heartbeat.

*Chico Hamilton, jazz drummer*

The literary scholars contributing to this volume take diverse approaches to metrical theory. The linguists, by contrast, work within a single tradition—generative metrics. Hence, a reasonable place for an introduction such as this to begin is with a brief historical survey of generative metrics and the controversy that it has caused among metrical theorists.

Traditional metrical theories define the IAMBIC PENTAMETER LINE as a sequence of five metrical feet, each composed of an unstressed syllable followed by a stressed one. Because poets frequently deviate from this strict pattern, traditional prosodists have found it necessary to add a list of permissible exceptions to their basic definition. For example, weakly stressed syllables may be permitted anywhere in a line; iambic feet may be replaced by trochaic or pyrrhic ones, or by spondees (heavy feet), or even by anapests; extrametrical syllables may be added to the end of the line, unstressed syllables omitted from the beginning of the line; and so on.

From the point of view of generative grammar, traditional descriptions such as this are unsatisfactory because they cannot distinguish metrical lines from unmetrical ones. Only four kinds of disyllabic feet are possible (iambic, trochaic, pyrrhic, and spondaic); hence, if all four are permitted in iambic

verse, then any sequence of ten syllables can be made to scan. Halle and Keyser (1966) propose to correct this deficiency in their seminal article on generative metrics, "Chaucer and the Study of Prosody":

Rather than look upon headless lines, lines with an inverted first foot, etc., as somehow deviant . . . we propose below a set of principles or rules which by their nature yield a large variety of metrical patterns, in the same way that rules of syntax yield a large variety of syntactic patterns. With respect to these rules, there will be one of two possible judgments. Either a line is metrical by virtue of conformity to the rules, or else a line is unmetrical by virtue of nonconformity to the rules. . . . [I]t is precisely this distinction which a theory of prosody of a given poet or poetic tradition must make. (Halle and Keyser 1966:371–372)

Relying primarily upon the metrical principles formulated by Otto Jespersen in "Notes on Metre" (1933), Halle and Keyser (1966) propose a generative definition of Chaucer's iambic pentameter line:

PRINCIPLE 1. The iambic pentameter verse consists of ten positions to which may be appended one or two extrametrical syllables.

PRINCIPLE 2. A position is normally occupied by a single syllable, but under certain conditions, it may be occupied by more than one syllable or by none.

Condition 1. Two vowels may constitute a single position, provided that they adjoin or are separated by a liquid or nasal or by a word boundary, which may be followed by *h*—, and provided that one of them is a weakly stressed or unstressed vowel.

Condition 2. An unstressed or weakly stressed monosyllabic word may constitute a single metrical position with a preceding stressed or unstressed syllable.

PRINCIPLE 3. A stress maximum may only occupy even positions within a verse, but not every even position need be so occupied.

Definition. A stress maximum is constituted by a syllable bearing linguistically determined stress that is greater than that of the two syllables adjacent to it in the same verse. (Halle and Keyser 1966:380–381)

Every aspect of this original definition has been disputed not only by traditionalists but also by generative metrists themselves. Controversy begins with the first sentence, which substitutes a sequence of ten metrical positions for the traditional five feet of iambic pentameter. Ironically, following Kiparsky (1977), generative metrists have tended to readopt the foot (as in the

articles by Hayes, Prince, and Youmans in this volume), whereas scholars from other traditions often dispense with foot boundaries (as in the articles by Tarlinskaja and Attridge).

Halle and Keyser's second principle is less controversial: everyone agrees that headless lines do occur occasionally in iambic verse and that two syllables sometimes substitute for one. In theory, however, Halle and Keyser's formulation permits a line of twenty-two syllables (two syllables per position plus two extrametrical syllables), while traditional metrics, which allows anapestic feet to be substituted for iambic ones, permits no more than seventeen syllables (five anapests plus two extrametrical syllables). Indeed, traditional metrists usually assume a limit of two, or at most three, anapests per line (implying an upper limit of fifteen syllables).

As generative metrists are inclined to readopt the foot, their account of disyllabic occupancy of single metrical positions tends to converge with the traditional view, although the specific phonological conditions permitting such occupancy have remained controversial. Halle and Keyser (1971:169) replace Conditions 1 and 2 with a single correspondence rule: "A position (S, W, or X) corresponds to a single syllable or to a sonorant sequence incorporating at most two vowels (immediately adjoining or separated by a sonorant consonant)." Kiparsky's article in this volume argues that syllable length, rather than the phonological boundary between syllables, is the determining factor: two syllables may occupy a single (weak) metrical position if and only if both are short. Edward Weismiller's article, in contrast, contends that generalized phonological rules are inadequate to explain poetic practice. Weismiller argues that pronunciation doublets existed in Renaissance England for a large number of words and phrases (some of which he lists in his appendix). Poets were free to include these words in either their expanded or their compressed forms in verse. (The radical compressions in Webster's plays are analyzed within the framework of generative metrics in Schlerman 1984.) By the nineteenth century, many of the words cited by Weismiller were no longer contracted in speech, but poets continued to follow the traditional practice of allowing such words to count as either shorter or longer sequences for metrical purposes. The result was to introduce triple rhythms into duple meter, and Weismiller believes that this relaxation was a contributing factor in the evolution of free verse.

The most distinctive (and most controversial) aspect of Halle and Keyser's theory is the Stress Maximum Principle (SMP): "A stress maximum may only occupy even positions within a verse, but not every even position need be so occupied." Halle and Keyser (1966) define a stress maximum as any syllable receiving greater stress than the two syllables adjacent to it in the same line of verse. (Because the first and last syllables in a line are adjacent to only one syllable, they cannot constitute stress maxima.)



Too many counterexamples to the original version of the SMP were discovered, so Halle and Keyser (1971:169) narrow the definition of the stress maximum to apply only to “fully stressed” syllables “between two unstressed syllables in the same syntactic constituent within a line of verse.” Minor stress peaks and peaks adjacent to phrase boundaries are no longer constrained categorically in the 1971 version of the SMP. To compensate for this weakening in the predictive power of the categorical component of their theory, Halle and Keyser strengthen its statistical component by establishing a tension metric, a formula for computing the metrical complexity of iambic lines. Least complex are lines with fully stressed syllables in all S positions and unstressed syllables in all W positions. More complex are lines with unstressed syllables in S positions and stressed syllables in W positions. Metrical complexity is presumed to correlate negatively with frequency of occurrence; that is, less complex lines tend to occur more often in verse than do more complex lines. Prohibited lines (with stress maxima in W positions) occur rarely, if at all.

An impressively large number of scholarly publications were written in response to Halle and Keyser’s seminal articles. Prosodists working within other theoretical traditions were quick to attack generative metrics, but generative metrists themselves were largely content with refining, verifying, and applying Halle and Keyser’s principles until Kiparsky (1975), whose work represents the first significant division within the generative camp itself. Kiparsky claims that the chief constitutive principle of the verse of poets such as Shakespeare is not the SMP at all, but the Monosyllabic Word Constraint (MWC). Briefly, the MWC requires all major stresses in W positions to be monosyllabic words unless these stresses are immediately preceded by phrase boundaries. In effect, the MWC is a constraint on polysyllabic words: major stresses in such words must align with S positions in verse (unless the misaligned stresses are preceded by phrase boundaries). The position of monosyllabic words and minor category words (such as articles, prepositions, conjunctions, auxiliary verbs) is not constrained at all by the MWC, nor are poets such as Milton, for whom the MWC is a normative, but not a categorical rule. Hence, the categorical component of Kiparsky (1975) has even less predictive content than that of Halle and Keyser (1971), and Kiparsky must rely on a tension metric to account for the metrical practice of poets in the many cases that are not governed by the MWC.

Just two years later, in 1977, Kiparsky proposes a major revision in his own theory. Motivated by the new theory of stress formulated by Liberman and Prince (1977), Kiparsky substitutes a binary, branching-tree notation for the linear, numerical representation of stress relations found in Chomsky and Halle (1968). Aided by this new notation, Kiparsky extends the MWC to apply

to secondary as well as to primary stresses within polysyllabic words and also (though less strictly) to stresses within clitic phrases. Kiparsky also readopts the foot as a constitutive metrical unit. Poets such as Milton, who are not bound by the MWC, allow polysyllabic words to straddle foot boundaries, and they occasionally allow stresses in polysyllabic words to occupy W metrical positions, but they do not allow both at the same time. That is, according to Kiparsky, iambic words such as *enough* and *divine* must occupy a WS foot in Milton, whereas trochaic words such as *future* and *surface* can occupy either SW or WS positions. Kiparsky also revises his tension metric. As before, complexity increases whenever stresses fail to match the prototypical WSWWSWSWS pattern, but tension also increases whenever word and phrases boundaries fail to coincide with foot boundaries.

In articles published in 1982, in 1983, and in this volume, I conclude that the MWC is more successful than the SMP in predicting the syntactic inversions that occur in Shakespeare's and Milton's poetry. Nearly half of these inversions have the effect of preventing violations of the MWC:

- (1) a. *Thy edge should blunter be than appetite* (Son. 56.2)  
 b. \**be blunter* (Violates the MWC)

Such inversions occur even when they are unnecessary to prevent violations of the SMP—indeed, even when they increase the number of mismatches between expected and actual stresses in the line:

- (2) a. *Under spread Ensigns marching might pass through* (PL 2.886)  
 1 2 3 4 5 6 7 8 9 10 (5 mismatches)  
 b. *Might pass through marching under spread \*ensigns*  
 1 2 3 4 5 6 7 8 9 10  
 (Violates the MWC)  
 (3 mismatches)

According to the tension metric of Halle and Keyser (1971), this line from *Paradise Lost* should be more complex metrically than its prose paraphrase. For Kiparsky, however, such lines merely illustrate the normative significance of the MWC.

Analysis of Shakespeare's and Milton's syntactic inversions demonstrates that the MWC is a strongly normative principle in their verse, and indeed, few metrical theorists—generative or traditional—would disagree with this claim. However, Kiparsky's constraint does not appear to be categorical even in Shakespeare, as sequences such as *enough noble* and *divine nature* in SWSW positions attest. I argue, therefore, that generative metrical rules such as the

MWC are statistically normative principles rather than categorical constraints. Under this view, the task of metrical theorists is to establish a metrical prototype and determine degrees of deviation from this prototype rather than to define a precise boundary between metrical and unmetrical lines. I claim that principles such as the SMP and the MWC should not be granted independent status as categorical rules; rather, they should be absorbed into a tension metric.

In his article in this volume, Bruce Hayes provides a generative analysis of the very regular trochaic meter of Longfellow's *Hiawatha*, and in doing so, he develops an unusually thorough and sophisticated generative theory of duple meter in English verse. Following Selkirk (1978, 1980, 1981), Hayes assumes that utterances are phrased hierarchically. At the lowest level are words, followed in ascending order by clitic groups, phonological phrases, intonational phrases, and utterances. According to Hayes, the meter in *Hiawatha* is constrained by a strict BOUNDING RULE: a stress peak defined at any prosodic level may occupy a metrical W position if and only if it is adjacent to another peak defined at the same prosodic level (where a peak is defined as any syllable with stress greater than at least one of its neighbors). For poets less strict than Longfellow, this principle is statistically normative, not categorical, but Hayes defines more permissive variants of the Bounding Rule that apply to other poets as well.

The Bounding Rule combines features of both the SMP and the MWC, and Hayes intends his rule (in its various forms) to displace these earlier principles. Indeed, with Hayes's article, generative metrists seem to have reached consensus on several points, most of which are likely to be acceptable to traditional prosodists:

1. Stress contrasts are more salient within words than between words, more salient within clitic groups than between clitic groups, and so on up the prosodic hierarchy. Consequently, metrical rules constrain the alignment of polysyllabic words most strictly, clitic phrases next most strictly, phonological phrases next most, and so on.

2. Atypical positioning of stressed syllables is less disruptive (and therefore more likely to occur) at the beginning of higher prosodic levels than at the beginning of lower ones. Thus, trochaic substitutions in iambic verse occur most often after utterance and intonation boundaries (especially at the beginning of lines of verse), next most often after phonological phrase boundaries, and so on down the phonological hierarchy.

3. Conversely, extrametrical syllables are more likely to occur at the end of higher-level prosodic constituents than at the end of lower-level ones.

There seems to be widespread agreement over these principles. In addition, I would propose the following revised version of a compensation principle

suggested in Hayes's article:

COMPENSATION PRINCIPLE. If a phonologically salient syllable occurs in a W position, then it is normally adjacent to another, more salient syllable in an S position.

This principle seems to be normative for all metrical verse. It is least normative for syllables at the beginning of a line or hemistich, where major phonological boundaries are most likely to occur. Elsewhere it is strongly normative, perhaps even obligatory for some poets—especially at the end of a line or hemistich.

Kiparsky and Hayes have amply demonstrated that phonological salience is not determined by stress alone. For example, the Compensation Principle predicts that a phrase such as *divine soul* is more likely to be positioned WSW rather than SWS because stress contrasts are more salient at lower prosodic levels than at higher ones; that is, the second syllable in *divine* is more salient for metrical purposes than *soul* is, even though the latter receives greater stress. Note that Shakespeare does position *divine soul* WSW in (3):

- (3) *Or my divine soul answer it in heaven* (R2 1.1.38)

By contrast, Shakespeare positions phrases such as *divine Nature* SWSW, since the stressed syllable in the second disyllabic word is more salient than that in the first:

- (4) *Thou divine Nature . . .* (Cym. 4.2.170)

Indeed most of the violations of the MWC that I have discovered in Shakespeare follow this pattern—a misaligned word such as *divine* is followed by an adjacent polysyllabic word that is aligned correctly:

- (5) a. *The **divine** Desdemona . . .* (Oth. 1.3.73)  
 b. *Thou **divine** Imogen . . .* (Cym. 2.1.62)  
 c. *Were **enough** noble . . .* (R2 4.1.118)  
 d. *Not **enough** barbarous . . .* (Per. 4.2.70)  
 e. *Has **deserved** prison . . .* (WT 2.1.120)

The Compensation Principle does seem to be operating in such lines. In any event, generative metrists now generally agree that measures of phonological salience must take the prosodic hierarchy into account as well as relative stress.

Traditional metrists can hardly object to this conclusion; English prosodists have long noted, for example, that polysyllabic words are constrained more strictly than monosyllables in verse. Indeed, one might claim that generative metrics is now moving toward a consensus that is largely consistent with traditional theories (the difference being largely in the greater detail and

formal rigor sought in the generative model). Why then has generative metrics aroused such hostility among metrists from other traditions? The reasons are due partly to the different aims and conventions of linguistics and literary scholarship. A clash between the two approaches became inevitable when Halle and Keyser chose to publish their first article on generative metrics in *College English* rather than in a linguistics journal. Not surprisingly, this article violated several conventions of literary scholarship, as have most articles written by generative metrists since. Reaction among literary scholars has been vigorous: generative metrists have been accused of ignorance, arrogance, and (worst of all) inaccuracy.

These charges are of course largely true, as they are for scholars in every discipline. Knowledge is necessarily finite and ignorance infinite in any field of study, and a degree of arrogance is required for anyone to put pen to page with the intention of instructing others. But from the point of view of literary scholars, generative metrists seem more ignorant and arrogant than usual partly because they follow an unfamiliar research model. Linguists, who are often required to describe exotic and unfamiliar languages as well as familiar ones, are accustomed to advancing highly provisional hypotheses based on limited data. These hypotheses are offered more as aids to research than as fixed beliefs. Indeed, an “interesting claim” in linguistics is one that is formally explicit, one that is easy to falsify (if incorrect), and one that provides the narrowest possible explanation for the available data.

Students of literature follow a very different research model: ideally, metrical theorists publish their pronouncements only after years of studying particular poets and poetic traditions and after assimilating the enormous body of literature about metrics that has accumulated for centuries. Attridge, Bjorklund, Tarlinskaja, and Weismiller work within this tradition. Weismiller’s article, for example, contains the statement, “There are virtually no genuine triple rhythms in English nondramatic syllabo-tonic verse before the Romantic period.” No linguist represented in this volume can speak with similar authority on the whole of English poetry, and scholars who have spent their entire careers studying English meter can be forgiven if they are inclined to view some of the “interesting claims” of generative metrists as uninformed falsehoods rather than as promising starting points for subsequent research. Nevertheless, as articles in this volume illustrate, early generative theories have by now evolved into sophisticated descriptions of meter—ones that are neither obviously false nor trivially true—and generative metrists have begun to incorporate more findings from other scholarly traditions into their work, along with insights from recent phonological theory. Hence, some of the old complaints against generative metrics apply with less force than they once did.

Another difficulty that literary scholars sometimes have with generative metrics arises from a misunderstanding of the term “generative” itself. This